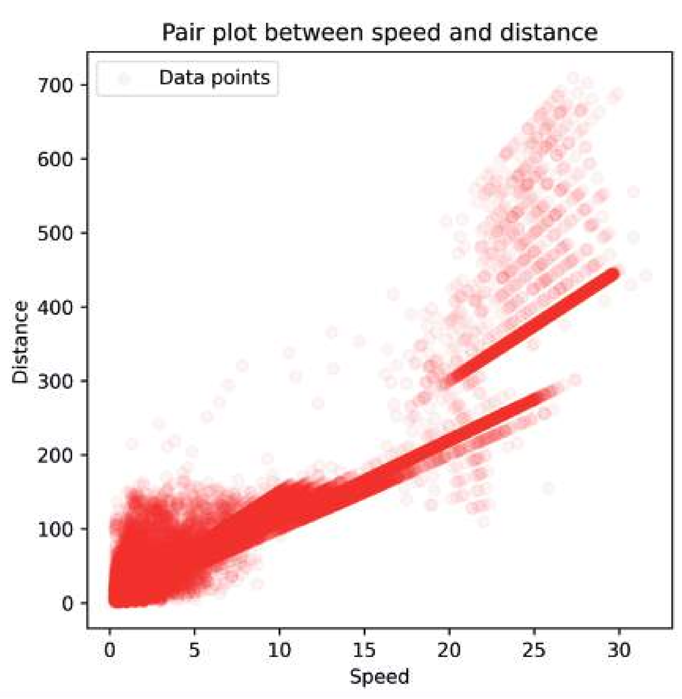
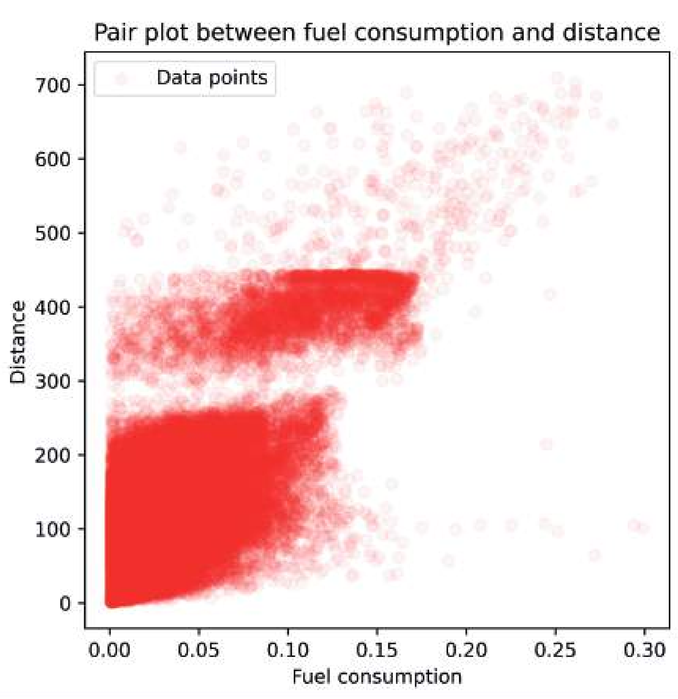
KATHOLISCHE UNIVERSITÄT EICHSTÄTT-INGOLSTADT

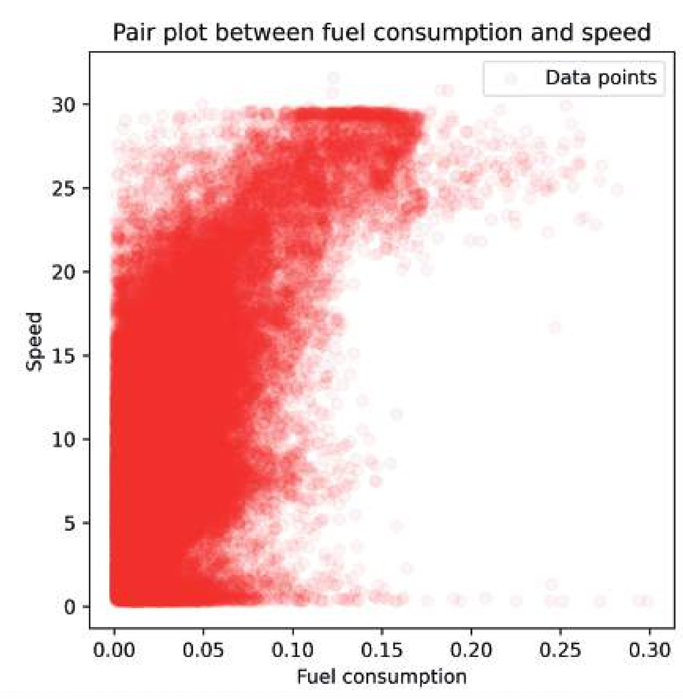
MIDS

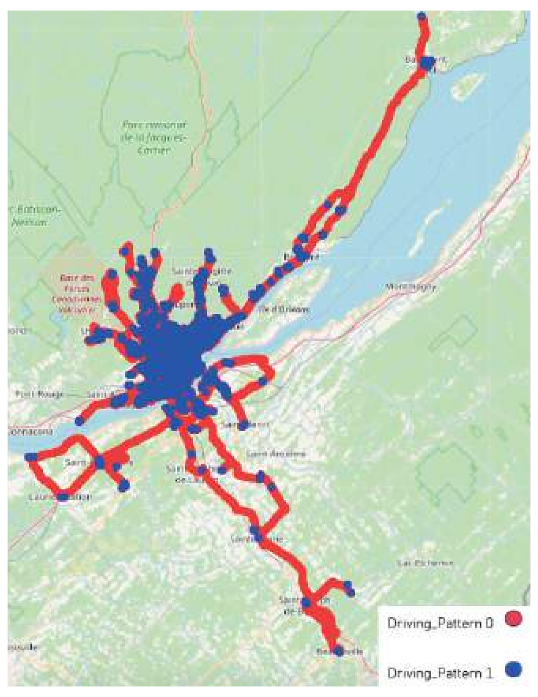
Hands-on Machine Learning and Data Science: ON ROAD FUEL CONSUMPTION

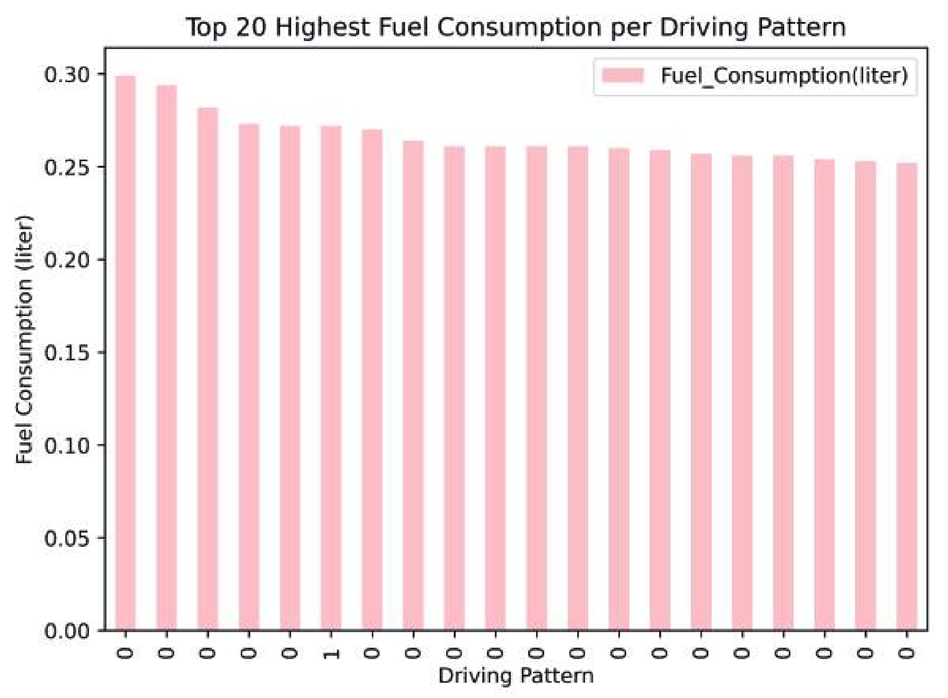
Explanatory Data Analysis (EDA)











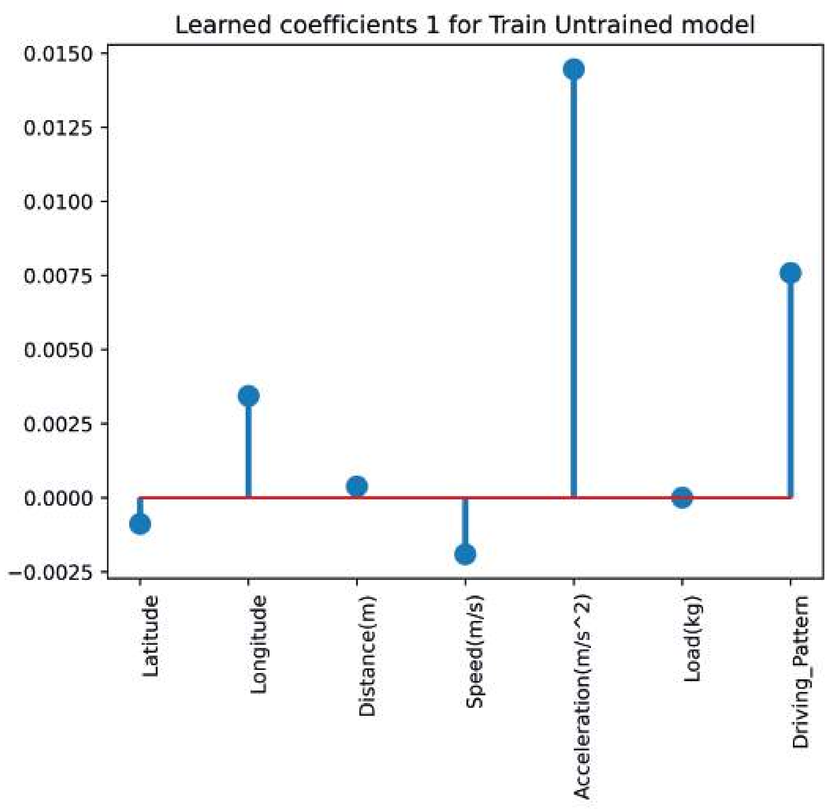
Predicting Fuel Consumption

Information about data

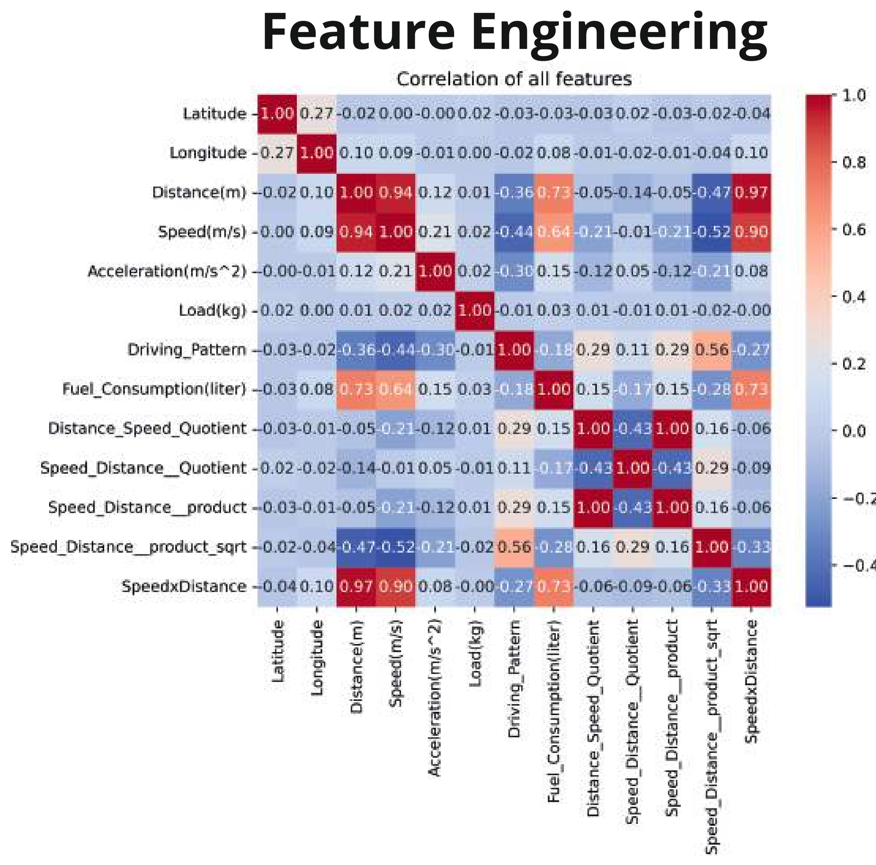
RangeIndex: 46476 entries, 0 to 46475
Data columns (total 8 columns):

#	Column	Non-Null Count	Dtype
0	Latitude	46476 non-null	float64
1	Longitude	46476 non-null	float64
2	Distance(m)	46476 non-null	int64
3	Speed(m/s)	46476 non-null	float64
4	Acceleration(m/s^2)	46476 non-null	float64
5	Load(kg)	46476 non-null	float64
6	Driving_Pattern	46476 non-null	int64
7	Fuel_Consumption(liter)	46476 non-null	float64

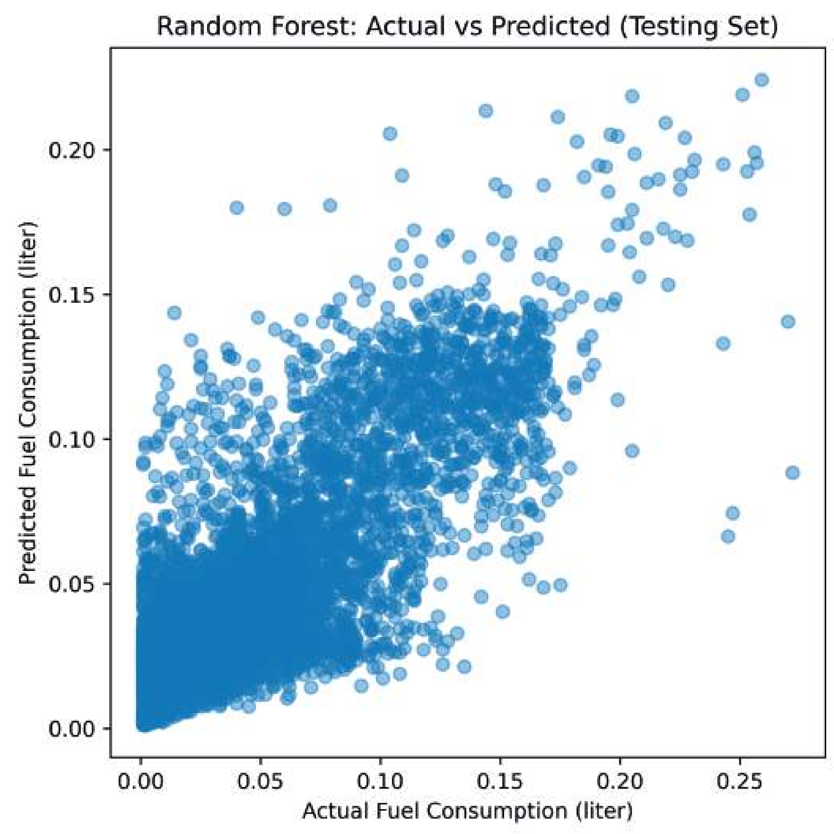
Coefficients plot without scaling



Correlation matrix/ Feature Engineering

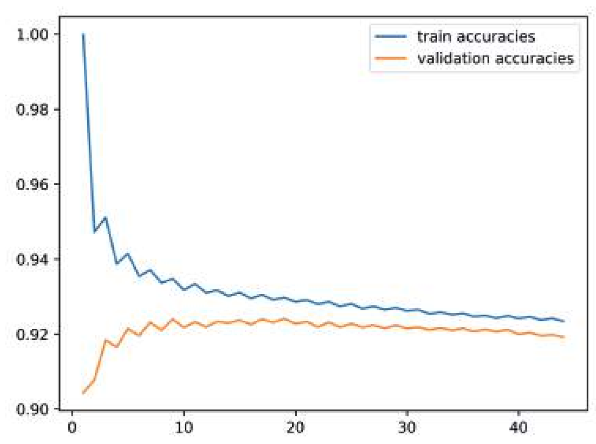


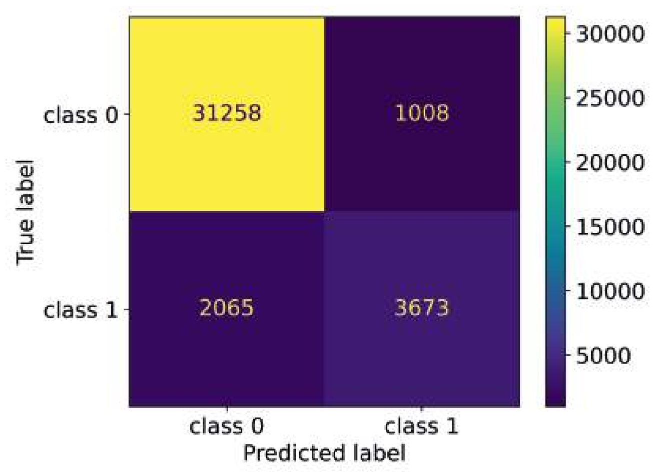
Model Prediction

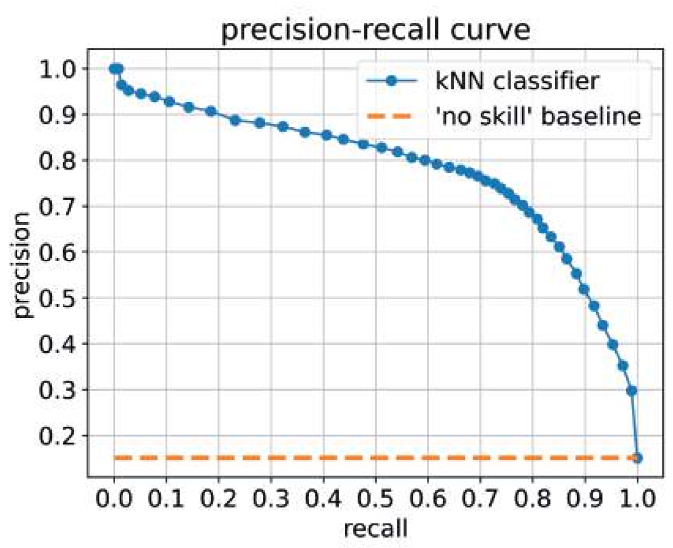


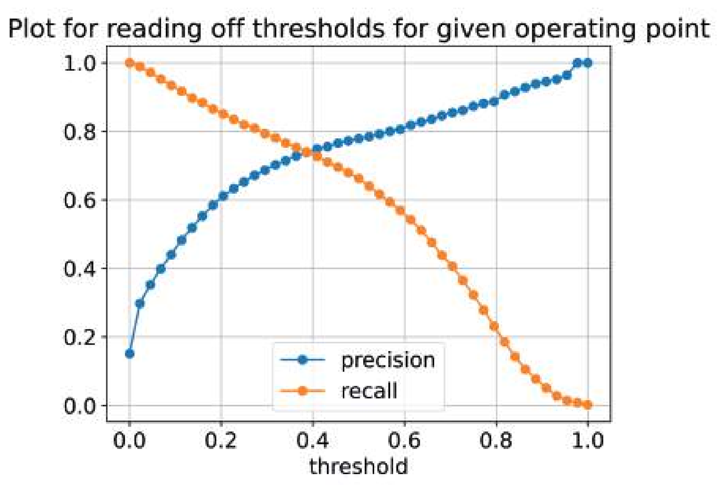
Predicting Driving Pattern

k-nearest neighbors

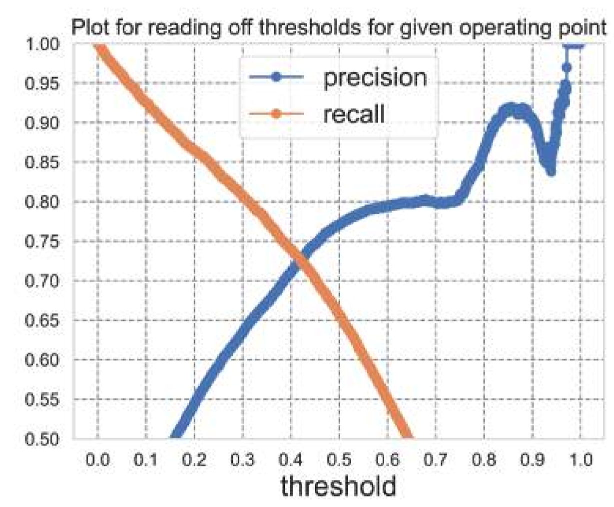


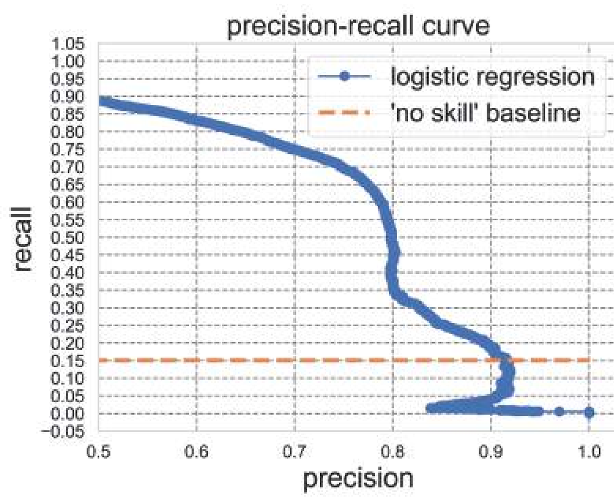




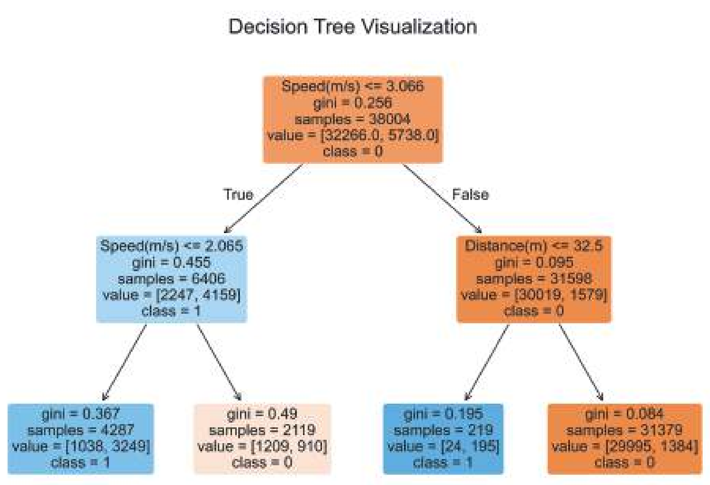


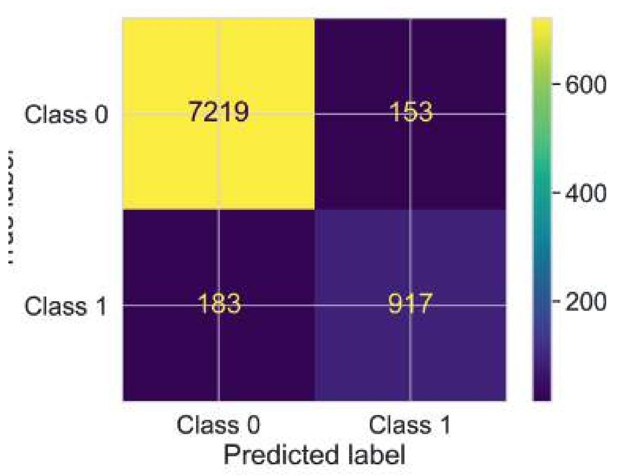
Logistic Regression



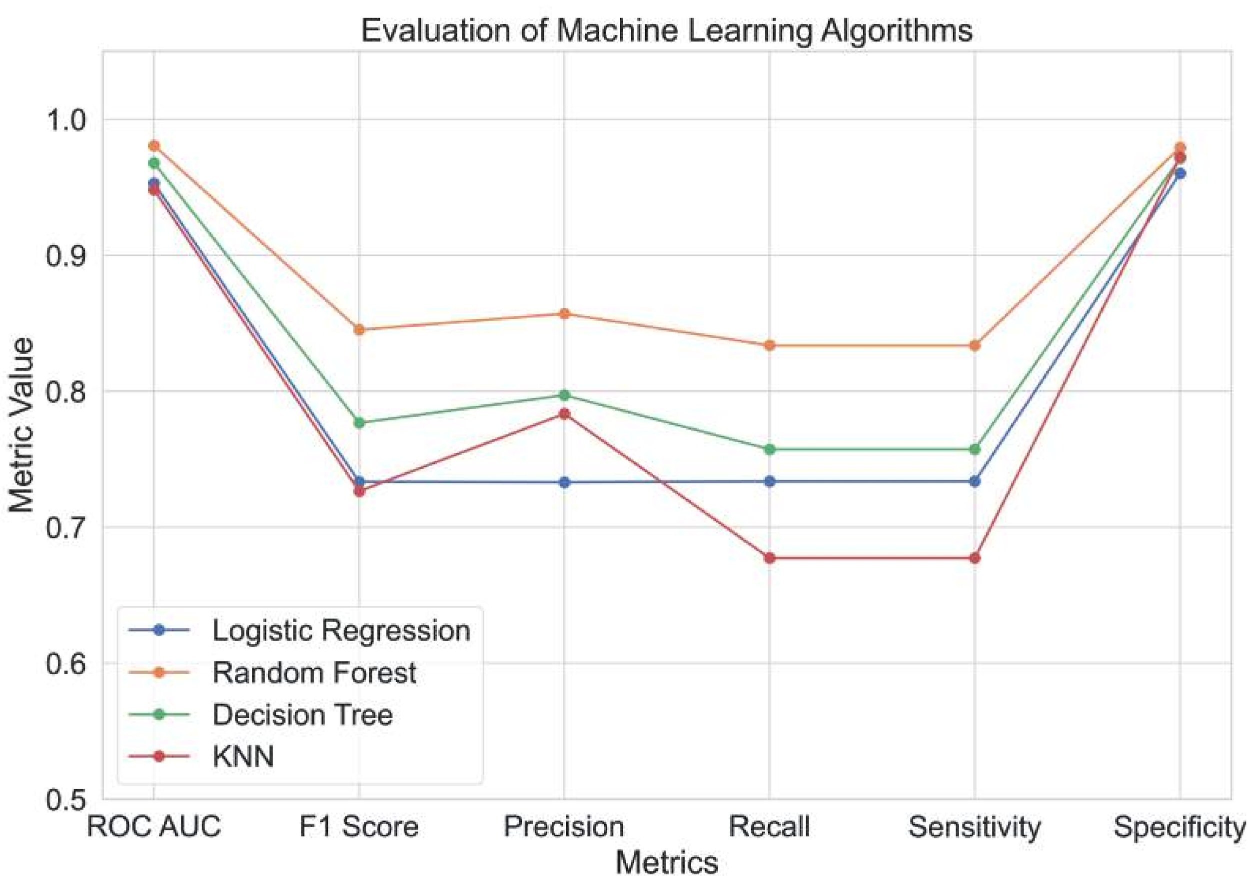


Decision Trees and Random Forests





Comparison of different models



This is how a tree would look like with default parameters without tuning it